A Classical Machine Learning TuberCulosis Comorbidity and Risk Factor Screening in India

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Certificate

This is to certify that the work contained in the project report entitled "**A Classical Machine Learning TuberCulosis Comorbidity and Risk Factor Screening in India**", submitted by **Raj Lakshmi (241723002)** for the award of the degree of M.Sc. Data Science to **Shree Guru Gobind Singh Tricentenary, Gurugram** is a record of bonafide research work carried out by her under my direct supervision and guidance.

The work presented here meets all the standards set by SGTU. The contents embodied in the project report are authentic and have not been submitted for the award of any other degree or diploma in this or any other university.

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Declaration

I **Raj Lakshmi**, hereby declare that the dissertation entitled "**A Classical Machine Learning TB Comorbidity and Risk Factor Screening in India**" submitting in partial fulfillment of the requirements for the award of the degree of Master of Science in Data Science at **Shree Guru Gobind Singh Tricentenary University**, is an original research work carried out by me under the supervision and guidance of my academic mentor.
I further declare that this work has not been submitted previously for any academic qualification, or any other institution or university for the award of any diploma, degree, or other academic qualification.

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This is to certify that the above statement made by me is correct to the best of my knowledge.

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Abstract

Tuberculosis (TB) continues to be one of the major health problems in India, with the country contributing around 28 percent of the total TB cases in the world. Treating TB becomes more difficult when patients also suffer from other health conditions such as Diabetes Mellitus (DM), tobacco addiction and alcohol misuse. These conditions can weaken the body, affect medicines, and reduces the chances of successful TB treatment. Under India's National TB Elimination Programme (NTEP), all TB patients are screened for diabetes, tobacco use, and alcohol consumption in both government and private hospitals. This creates a large amount of state-level healthcare data that tracks three important stages: screening of patients, identification of the disease or addiction, and referral or treatment support provided to the patient. This dissertation studies the NTEP data from all 36 states and Union Territories of India using data analysis and machine learning techniques. The work includes collecting and cleaning the data, handling missing values, and applying statistical and predictive models to understand healthcare performance across states. Different methods such as correlation analysis, clustering models, and forecasting techniques were used to identify patterns and predict future treatment outcomes. The study found that India performs relatively well in the first stage of screening. Around 91.7 per cent of TB patients were screened for diabetes, while 81.7 per cent and 80.9 per cent were screened for tobacco and alcohol use, respectively. However, a major problem was identified in the final stage of treatment support and referral. Only 63.3 per cent of TB patients diagnosed with diabetes received anti-diabetic treatment. The situation was worse for tobacco and alcohol users, where only 31.9 per cent and 33.5 per cent of identified patients were referred to Tobacco Cessation Centres and De-Addiction Centres. The analysis further showed that these referral gaps exist in both government and private healthcare systems, which indicates that the issue is not limited to one sector but reflects a broader weakness in healthcare infrastructure and patient support systems.
Overall, this dissertation highlights how data science and machine learning can help identify weaknesses in public healthcare systems and support better decision-making. The study also demonstrates that machine learning models can successfully classify states based on healthcare performance and predict treatment outcomes with reasonable accuracy.

List of Abbrevations

S.No.	List of Abbrevations
WHO	World Health Organization
TB	Tuberculosis
DM	Diabetes Mellitus
NTEP	National TB Elimination Programme
NCD	non-communicable disease
TCCs	Tobacco Cessation Centres
IQR	Interquartile range
PCA	Principal Component Analysis
DAC	De-Addiction Centre
EDA	Exploratory Data Analysis
LOO-CV	Leave-One-Out Cross-Validation

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Chapter 1

Introduction

In human history, Tuberculosis is one of the oldest and most dedicated infectious diseases globally. In the twenty-first century, it is one of the highest causes of morbidity and death worldwide [1]says that the TB deaths are caused by an estimated 1.3 million in globally, and India is bearing the heaviest burden of the single nation. Worldwide, there are 27 to 28 per-cent new TB cases. This isn't merely burden reflection of the population size of India, it is also a result of complicated socio-economic, nutritive, and epidemiological provisions that nourish transmission and obstruct prosperous treatment.

Over the past decades, a growing body of clinical evidence and epidemiological has determine that TB is rarely exists in clinical isolation. The increasing proportion of TB patients are present concurrently with one by one and non-communicable disease (NCD). Through these, in Indian context the particular programmatic importance are: usage of tobacco,misuse of alcohol, Diabetes Mellitus (DM). These conditions operates among behavioural mechanisms and interacting biological TB epidemiology. It increases the risk of relapse, treatment failure and drug resistance due to active disease, sensitivity of infection.

These three co-morbidities generates that public health refer to as a 'syndemic' - a set of mutually epidemics, co-occurring that share social determinants which require fused rather than responses of vertical programmatic. In India, the burden of type 2 diabetes are rising with an estimation of 101 million individuals while 2023, and tobacco usage is over 267 million and the major rates of risky alcohol consumption, whether India can achieve the aspiring quarry set by its NTEP.

Every TB patient should be screened for DM, tobacco use and alcohol use and notified to NTEP through the NIKSHAY digital platform. If they found positive, then it must be linked to the suitable treatment and services such as anti-diabetic medicine for TB diabetes patients, Tobacco Cessation Centres(TCCs) used for tobacco users, and De-Addiction Centres (DACs) for dangerous alcohol users.

For the purpose of treatment, the existence of both DM and TB creates a clinical management challenges. The first-line of anti-tuberculosis chemotherapy is Rifampicin, it is a potent persuader of hepatic cytochrome P450 enzymes and it also reduces plasma

concentrations of hypoglycaemic agents by $20 - 40\%$, and give standard doses of anti-diabetic medication during TB treatment. Again, TB infection causes a inflammatory state that decline insulin resistance. Studies states that patients having TB with DM have 69% high risk of treatment failure, the default risk is 89%, the higher risk of relapse is 389%, and mostly higher death linked to TB patients without diabetes is making effective DM management during TB management and not merely a chronic issue but a direct determinant of TB programme outcomes.

1.1 TB-Comorbidity Nexus in India

In scientific literature, states that the relation between the three primary comorbidities and TB is well established, but in the Indian context, it deserves careful examination. Every comorbidity is handled through a different aetiological pathway, and simultaneously, they make a compounding burden that generally elevates the hardship of TB management.

1.1.1 Diabetes Mellitus and Tuberculosis

In clinical medicine, diabetes and tuberculosis are the most highly documented in comorbid relationships. The individuals having diabetes contains one to three-fold higher risk of growing active TB equated to normoglycaemic individuals. The well-characterised immunological mechanisms are: the cellular defence of Mycobacterium tuberculosis, which reduces the output of oxygen species and anti-bacterial cytokins, and with the help of T-helper cells, it compromises the adaptive immune system. At the same time, insulin resistance pairs with glycaemic control among inflammation of insulin signalling pathways.
India is the second-largest country where people are living with diabetes, and the number of individuals at risk is increasing day to day. Past studies have shown that some countries like Andhra Pradesh, Tamil Nadu, South India, have found that 15 to 25 per cent of TB patients are positive for diabetes, and it's larger than the global average of 10 to 15 per cent. Likewise, TB patients with diabetes have large sputum conversion times, rates of treatment failure are high, the risk of relapse and drug resistance is elevated.

1.1.2 Tobacco use and Tuberculosis

Tobacco use and Tuberculosis have relation between each other which operates among respiratory damage and immune respiratory. Cigarette holds smoke of more than 7000 chemical compounds, most of them are directly toxic to the respiratory system.
The epidemiological risk is substantial. The current smokers have approximately 2.3 times the risk of active patients of TB compared to non-smokers. Bidi smokers face

risks comparatively to cigarette smokers in India. Smokeless tobacco users show elevated TB risk, however the magnitudes are less characterised.

1.1.3 Alcohol Misuse and Tuberculosis

Alcohol use chaos and risky drinking interact with TB among a combination of immunological damage, and harms the lungs.
With the availability of this data, it has a quantitative analysis of NTEP's co-morbidity performance with the use of strict DS and ML models. In India the published literature of TB co-morbidities is mostly clinical or epidemiological in nature, main focus on biological mechanisms, programmatic description, or clinical trial outcomes. Use of ML analysis with state-wise dataset- explores how states perform differently, what factors can predict success, how both public and private sectors compare, and what path states are probable to follow. The gap between this represents significant scientific limitation and policy opportunity.

1.2 Problem Statement

The Screening co-morbidities infrastructure of India's NTEP has invested substantially in around 36 states and Union Territories. Though , the programme which exist doesn't give guarantee in effective delivery. The data is analysed in dissertation reveals a concerning and consistent pattern: a bit screening rates at Stage-1 (known status) are high in most states, later stages of the cascade, similarly Stage-3 (treatment initiation) shows lower performance. This cascade represents a failure with clinical consequences: Treatment outcomes go unmanaged due to TB patients who has co-morbidities, higher morality, and high risk of relapse.
This study motivates some precise research problems. Specific questions have not been answered in the published literature. Nobody has systematically applied machine learning to classify Indian states by their TB-comorbidity programme performance performance and use that classification to guide policy. Nobody has build a probabilistic forecast of India's probable comorbidity programme path for the next four years, landed in actual data. This dissertation tries both.

1.3 What This Study Aims to Do

The objective are not a formal list. Each one is corresponding to a concrete analytical step and it is described, executed and interpreted in chapters that follow.

1. Acquire, clean, and preprocess the TB dataset that makes every decision specific and justified.

2. Mapping and identify precisely that where and how the treatment of diabetes, tobacco, and alcohol are cascade at the national level and over all 36 states.

3. Comparing Public and Private sector performance using statistical testing, and decide the gap is actual or not.

4. Test whether performance through all three co-morbidities is correlated- regardless states that do well on diabetes will also do well in tobacco and alcohol and interpret that what means programmatically.

5. Use K-Means Clustering and PCA to group states into meaningful performance tiers that policymakers can actually act on.

6. Identify, through Random Forest and Gradient Boosting models, which features of a state's current performance best predict its overall comorbidity management quality.

7. Translate all of this into precise, evidence-anchored recommendations that are pointed adequately to be useful.

1.4 Research Gap

Existing studies lack machine learning-based state-wise analysis of TB-Diabetes interaction. The case of doing this research is not complicated. Tuberculosis is killing more people in India than it should, and comorbidities are a major part of why. A diabetes patient goes unmanaged while they are being treated for TB has a meaningfully worse chance of recovering. A patient who keeps smoking through TB treatment has worse sputum conversion rates and higher relapse risk. A patient who is drinking heavily is less probable to take medication correctly. These are not a theoretical statements; they are backed by years of clinical evidence.
NTEP's response mandates that screening and linkage for all the three comorbidities is the right tendency. But the data shows that the programme is not delivering on that instinct at anything close to the required scale. This dissertation uses DS to say, as just as possible, exactly how far of the mark we are, which states needs the severly help, and what the next four years might look like.
Despite the programmatic significance of TB-comorbidity management and the richness of the NIKSHAY data infrastructure, systematic data science analysis of state-level comorbidity cascade performance across all three NTEP-mandated comorbidities simultaneously remains absent from the published literature. Existing analyses either focus on single comorbidities, use clinical rather than programmatic data, examine individual states rather than the national picture, or report descriptive statistics without the inferential and predictive analyses that would generate actionable policy insights. This dissertation fills this gap through a comprehensive, methodologically rigorous data

science investigation employing the full analytical pipeline from data cleaning through machine learning to probabilistic forecasting.

1.5 Scope, Limitations, Expectations

In this study, it covers all 36 Indian states and UT's using NTEP co-morbidity data from 2024. It does not cover HIV-TB co-morbidity, which is managed under a separate PLHIV programme. The single most significant limitation is this: There is only one year of data available for quantitative analysis. That means conventional time-series forecasting models such as **ARIMA**, **LSTM**, and **Prophet** can't be used here without being technically fradulet, because they require substantially more historical time points than one. The forecasting chapter explains what we do instead, why the approach is defensible, but the limitation should be stated clearly up front.

Chapter 2

Literature Review

Tuberculosis in India: India's relationship with tuberculosis is long, painful, and not yet resolved. The disease has been documented on the subcontinent for millenia, but the modern public health response dates to early 1960's when the first national TB programme was established. That program evolved, through several restructuring, into the Revised National TB Control Programme (RNTCP) in 1997, which adopted the DOTS (Directly Observed Treatment, Short-course) strategy and genuinley Tuberculosis and Diabetes comorbidity has been widely evolved treatment success rates above the following decade [33]. In 2020, the RNTCP was rebranded as the National TB Elimination Programme (NTEP), with an ambitious stated target: eliminating TB in India by 2025, five years ahead of the WHO's global deadline [4].

The numbers suggests that target is not going to be met comfortably. The WHO Global TB Report for 2023 estimates that 2.8 million new cases in India in 2022, at the rate of 199 per 1,00,000 population and nearly 50% of above the global average of 133 per 1,00,000 [1]. TB mortality in India that same year was estimated at 3,42,000 and one quarter of all the TB deaths in worldwide. Detection of case has improved and reached approximately 80% in 2022, but still means one in five cases goes detected. Treatment success for drug-sensitive TB is around 85, which sounds reasonable until one considers that 15 failure represents hundreds and thousands of people per year [3].

The epidemiological picture is shaped heavily by social determinants. States with high poverty rates, dense urban housing, pour nutrition, and limited healthcare access have consistently worse TB outcomes [2]. Bihar, Madhya Pradesh, Uttar Pradesh, and Jharkhand are the states that appear most often in the bottom quartile of TB indicators, and, as this dissertation's own analysis confirms, they appear in the bottom cluster for comorbidity management as well. Private healthcare sector handles the majority of TB patients in many Indian states, and it has historically been difficult to integrate into accountability systems and programmatic monitoring [34]. In healthcare, ML has been applied in disease prediction, but in TB-DM integration the existence of work is very limited.

Tuberculosis has shaped human history in ways that few infectious diseases can match. It decimated European populations during the Industrial Revolution, inspired literary

and artistic movements through its association with romantic suffering, and provoked the first organised international public health response in the form of the International Union Against Tuberculosis and Lung Disease, founded in 1920. In the mid-twentieth century, the advent of effective chemotherapy- streptomycin (1943), isoniazid (1952), and rifampicin (1963)- appeared to herald the disease's imminent defeat. Yet TB proved more resilient than optimistic epidemiologists of the 1960s had predicted. The collapse of public health infrastructure in many post-colonial states, the devastating synergy with the HIV epidemic beginning in the 1980s, the emergence of multidrug-resistant (MDR) and extensively drug-resistant (XDR) strains of TuberCulosis, and the global rise of non-communicable disease risk factors- particularly diabetes have ensured TB's persistence as a major global killer.

2.1 When TB confronts Diabetes

Diabetes mellitus weakens the immune system in many ways that are particular dangerous for TB. The neutrophils and macrophages that are supposed to engulf and kill *Mycobacterium tuberculosis* function poorly in a hyperglycaemic environment. The immune system's method walling off TB infection is impaired. Cytokine production is essential for adaptive immunity against mycobacteria, are reduced [10, 11]. The consequence is a body that is significantly less capable of containing or cleaning TB infection. The relationships run between other directions too. Active TB infection causes stress hyperglycaemia and increases costisol production, which drives insulin resistance. Some anti-TB drugs are isoniazid and rifampicin and have metabolic effects that complicate glycaemic control . So a patient who is diagnosed with both the conditions simultaneously but diseases are actively making each other worse [9].

2.2 When TB Meets Tobacco

TB damages the person whom taking tobacco and it introduces more than 4,000 chemical compounds into respiratory track and many of them are well-documented effects on lung's ability to defend against inhaled pathogens. Mucociliary clearance i.e., the mechanical process which the airways sweep pathogens outward and its progressively impaired by tobacco smoke exposure. Alveolar macrophages, the immune cells that are the firt line of response to inhaled, these are both numerically reduced and functionally impaired in smokers. The tight junctions of the respiratory epithelium are loosened, making it easier for bacteria to establish a foothold [17, 18].
Lienhardt and colleagues published a rigorous systematic review that is still widely cited: tobacco smoking confers a 2.5-fold increased risk of TB infection, a 2.3-fold increased risk of progressing to active disease, and nearby three-fold increase in TB mortality [19]. In India, where approximately 267 million adults use tobacco in some

form (the GATS India survery makes this uncomfortably clear [20]), Bates and colleagues estimated that roughly 23 of all Indian TB cases are attributable to tobacco use [6]. Jha and colleagues, in a large New England Journal paper, reinforced the magnitude of the hazard across the Indian smoking population [21].
Beyond infection risk, tobacco has significantly worse treatment outcomes. Thomas and colleagues, in a cohort study from South India, documented that tobacco users had worse conversion rates at two months, lower overall treatment successm and has higher relapse rates compared to non-users [22].
The NTEP's response is to mandate referral to Tobacco Cessation Centres (TCCs) for all identified tobacco-using TB patients. Bam and colleagues reviewed tobacco cessation services at NTEP facilities and found that counselling was offered only in a minority of sites, and formal TCC referral was even less common [24]. The analysis in this dissertation confirms a national TCC linkage rate of 31.9% identified tobacco-using patients receive no formal cessation support.

2.3 Treatment Cascade in Public Health

Sometimes the treatment cascade called the care cascade or diagnostic-treatment cascade. It is an analytical framework originating in HIV/AIDS epidemiology that has been widely adopted across multiple disease areas for its ability to reveal specific programmatic bottlenecks in patient management pathways. The framework was most influentially operationalised through UNAIDS's ̈90-90-90̈targets, introduced in 2014, which called for 90% of HIV-positive individuals to be diagnosed, 90% of those diagnosed to be on antiretroviral treatment, and 90% of those on treatment to achieve viral suppression- creating a cascade with a clear numerically defined goal at each stage. The simplicity and clarity of framework, and the dramatic improvements in HIV programme performance that followed its adoption, prompted rapid translation to other disease contexts.
In the tuberculosis context, cascade frameworks were first applied to the diagnostic-to-treatment pathway anad characterising what proportion of estimated TB cases are detected, what proportion of detected cases are notified, what proportion are initiated on treatment, and what proportion achieve treatment success. This cascade became central to NTEP programme evaluation and drove investments in active case finding, diagnostic expansion (including the rollout of Truenat and GeneXpert molecular diagnostics), and treatment support interventions. The NTEP's extension of the cascade concept to comorbidity management adds a three-stage cascade for diabetes, tobacco, and alcohol screening, identification, and linkage. It represents a natural evolution of this framework to encompass the full clinical needs of TB patients.

2.4 Data Science and Machine Learning in Public Health Research

The application of data science methods to public health has accelerated dramatically over the past decade, driven by digitisation of health records, expansion of national health information systems like NIKSHAY, and growing availability of open-source computational tools. Within tuberculosis research specifically, machine learning has been applied to: automated TB detection from chest X-rays using convolutional neural networks (with accuracy approaching radiologist performance in some settings); prediction of MDR-TB from genomic data; identification of missed TB cases from primary care electronic health records; and spatial prediction of high-incidence areas for targeted active case finding.

A persistent methodological challenge in applying machine learning to national health system data from low and middle income countries is the small problem: state-level analyses typically yield 30–50 observations, insufficient for many standard ML approaches. This challenge has driven methodological innovation including the adoption of Leave-One-Out Cross-Validation for performance estimation in small-n settings, regularisation techniques to prevent overfitting (Ridge, Lasso, Elastic Net regularisation in regression; maximum depth and minimum leaf-size constraints in tree-based methods), and the use of bootstrap and Monte Carlo methods to quantify prediction uncertainty. This dissertation contributes to this methodological literature by demonstrating the appropriate application of these adaptations to a 36-observation national health system dataset.

2.5 Prior Studies on TB Comorbidity Management in India

Several prior studies have examined specific aspects of TB-comorbidity management in Indian states, though none has applied the comprehensive multi-comorbidity, multi-method analytical approach of this dissertation. [45,47] established the epidemiological foundation for DM-TB co-prevalence in South India. [48] synthesised evidence on the impact of DM on TB treatment outcomes across Indian studies, finding consistently elevated rates of treatment failure and default that have since informed NTEP guidelines. [49] provided a global synthesis of the DM-TB management challenge that explicitly identified India as a country requiring urgent programmatic response.

[50] made the most directly methodologically relevant prior contribution, applying Random Forest to patient-level NIKSHAY data to predict TB treatment outcomes, finding that alcohol use and diabetes were among the strongest predictors of unfavourable outcomes. Their demonstration that ML can extract actionable predictive information from NIKSHAY programmatic data provided both inspiration and methodolog-

ical precedent for the analytical approach in this dissertation. [51] applied K-Means clustering to district-level NTEP performance data, identifying high-priority districts for notification improvement- a methodological parallel to the state-level clustering approach employed here.

Chapter 3

Methodology

3.1 Dataset Description

The dataset consists of Three worksheets each one of them has 37 rows (36 states/UTS plus a national India aggregate row) and 19 columns. Column one has state name. The remaining 18 columns capture a six-metric cascade that splits into counts (N) and percentages (%) among three sectors: Public, Private, and Total. Table 3.1 summarize this.

Stage 1	Known Status	Public / Private / Total	TB patient been screened for the comorbidity? (No. and %)
Stage 2	Diagnosed / Identified	Public / Private / Total	Was a comorbidity actually found? (No. and %)
Stage 3	Treatment / Linked	Public / Private / Total	Did the patient receive treatment or get linked to a service? (No. and %)

Table 3.1: Dataset structure: stages, metrics, and column organisation

The India aggregate row was removed before all state-level analysis to prevent it from distorting descriptive statistics, correlations, and cluster assignments. Percentage columns were used as the primary analytical variables rather than raw counts, because counts are dominated by large states (Uttar Pradesh has roughly 5,00,000 notified TB patients; Lakshadweep has roughly 500). Percentages allow meaningful comparisons across this huge range.

3.2 Cleaning the Data

First of all, we have to clean the raw data to find the insights and to clean the data. There are some steps of cleaning:

3.2.1 Column Naming

The raw column headers in each worksheet were inconsistent and average roughly length was of 85 characters, with varied puntuation and terminology. Every column was renamed to a compact systematic format: (known, diagnosed/identified, treated/linked), `Sector` {Pub, Pvt, Tot} means: the total percentage of identified tobacco-using TB patients who are linked to a TCC. This naming convention made all code clean and readable.

3.2.2 Missing Values: What They Are and What To Do About Them

Across the three worksheets, 35 missing values were found. One of them was in a private-sector column, and every one corresponded to a small Union Territory such as Lakshadweep, Dadra, Ladakh, and Nagar Haveli and Daman and Diu, Andaman and Nicobar Islands (partial), or Arunachal Pradesh. This is not a random pattern. These UTs have negligible or zero private TB notification infrastructure, which is why no private-sector data was submitted. The missingness is Missing Not At Random (MNAR) which is structurally determined by the healthcare system, not by chance.
By Giving this three imputation options were considered. Deleting rows would remove genuine geographic diversity from an already small sample ($n = 36$). Mean imputation would be pulled by outliers that the Shapiro-Wilk tests later confirmed are present. **Median Imputation** was chosen: it preserves central tendency, is robust to the skewness confirmed in the data, and is entirely appropriate given that only 1.75% of numeric cells were missing.

3.3 Feature Engineering

After cleaning, five composite features were created:

- `Cascade_D`: mean of `D_KnownTotalperf`, `D_DxTotalperf`, `D_TxTotalPerf`- a single number of summarising diabetes cascade performance.
- `Cascade_T`: same logic for tobacco.
- `Cascade_A`: same logic for alcohol.
- `Overall_Comorbidity_Score`: mean of three cascade composites i.e. one number capturing overall programme quality for a given state. This is the target variable an all supervised ML models.
- `Pub_vs_Pvt_Gap`: `D_KnowPubPerf` − `D_KnowPvtPerf` - the public minus private sector screening gap for diabetes.

The Overall Comorbidity Score serves as the primary outcome variable for Random Forest regression and the principal target for predictive modelling in Chapter Six. It provides a single summary metric capturing a state's performance across the entire comorbidity management portfolio though its construction as a simple arithmetic mean across stages with different denominators is acknowledged as a simplification, and all stage-level analyses are also presented separately.

3.4 Statistical Methods

Standard descriptive statistics are compute for all key percentage variables: mean, median, standard deviation, interquartile range (IQR), minimum, maximum, and skewness. The emphasis on median alongside mean throughout the analysis reflects the non-normal distributions of several metrics, where the median provides a more robust measure of central tendency.

3.5 Exploratory Data Analysis

Univariate Analysis and Normality Testing are descriptive statistics which is computed for all 18 percentage variables per worksheet: mean, median, standard deviation, minimum, maximum, IQR, skewness, and kurtosis. Box plots were generated for each metric, stratefied by (public, private, total). Most unfavourably, the **Paired Wilcoxon Signed-Rank Test** is used to compare the variables over the TB datasets which involves diabetes, tobacco and alcohol.
Wilcoxon Signed-Rank Test: It is used to find the comparison between two dependent or paired samples when the assumptions of normality required for the pared t-test are not satisfied.
This test evaluates whether the median difference between paired observations is significantly different from zero. Different parametric methods, it does not assume that

data follows a normal distribution, and making it specific suitable for small and non-normally distributed healthcare datasets.
For this dataset, Wilcoxon test was appropriate because: Dataset contain small number of observations i.e. state-wise data. Healthcare indicators such as diabetes occurence, tobacco use, and alcohol use often exhibit skewed distributions and outliers. The variables collected from same states or regions.
The paired Wilcoxon test is using to compare:

- Diabetes-related TB indicators
- Tobacco-related TB indicators
- Alcohol-related TB indicators

3.6 Model Selection

The following models were used:

- K-Means Clustering
- Random Forest
- Principal Component Analysis (PCA)

Chapter 4

Results And Discussion

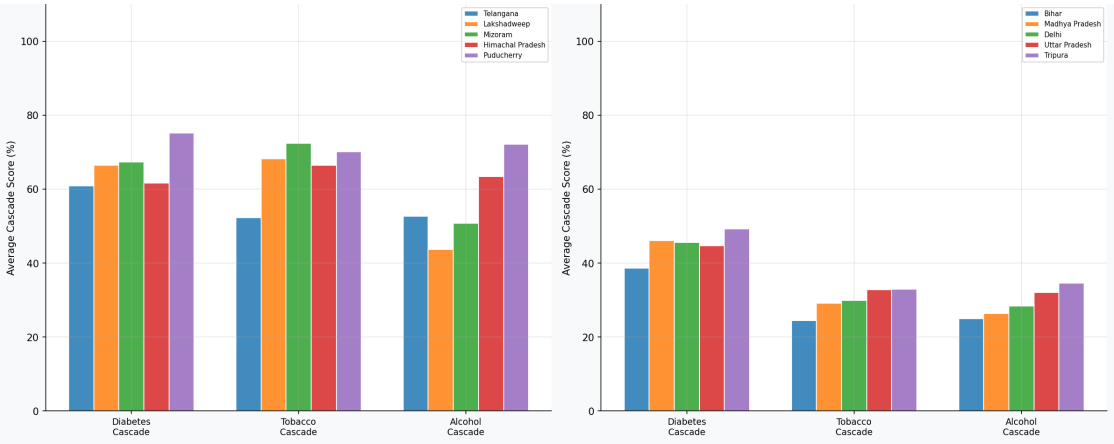


Figure 4.1: Top 5 vs Bottom 5 states Treatment Cascade

4.1 The Treatment Cascade

The most important result I'm finding of this analysis is Stage 3 i.e. treatment rates are dramatically lower than the Stage 1 i.e. screening rates across all three comorbidities, in both public and private sectors.

4.1.1 Overall Interpretation of the Cascade

The graph shows a common public-health pattern:

High Screening → Low Diagnosis → Moderate Treatment
This means: The cascade analysis indicates that, in the case of diabetes, India is performing relatively well in screening among TB patients, with 91.5% of patients having

SNo.	Known Status%	Diagnosed%	Treatment%
Diabetes	91.5%	7.8%	63.3%
Tobacco	75.5%	11.5%	31.9%
Alcohol	74.2%	7.2%	33.5%

Table 4.1: India-level treatment cascade: Performance at each stage

their diabetes status recorded. However, only 7.8% of TB patients were diagnosed with diabetes. Among the identified diabetic TB patients, 63.3% were linked to treatment, which represents the strongest treatment linkage among all three cascades. This suggests that diabetes management within TB programs is comparatively stronger than the management of behavioral risk factors. The findings indicate better integration of diabetes screening and treatment services within the national TB control framework.

In the case of tobacco, the tobacco status of TB patients was recorded for only 75.5% of patients, indicating that tobacco screening is weaker than diabetes screening. Among the screened individuals, 11.5% were identified as tobacco users. However, only 31.9% of these patients received cessation treatment linkage, which represents a major public-health concern. The results suggest that India is relatively better at identifying tobacco use than providing effective treatment and cessation support. This may reflect the lack of adequate de-addiction infrastructure, poor behavioral intervention systems, and weak follow-up mechanisms within healthcare programs.

For alcohol-related comorbidity, the screening coverage was the lowest among all cascades at 74.2%. This suggests that alcohol-use history is less consistently recorded among TB patients. The identification rate of alcohol-use risk was only 7.2%, which may be influenced by underreporting, social stigma, or incomplete disclosure by patients during screening and data collection procedures.

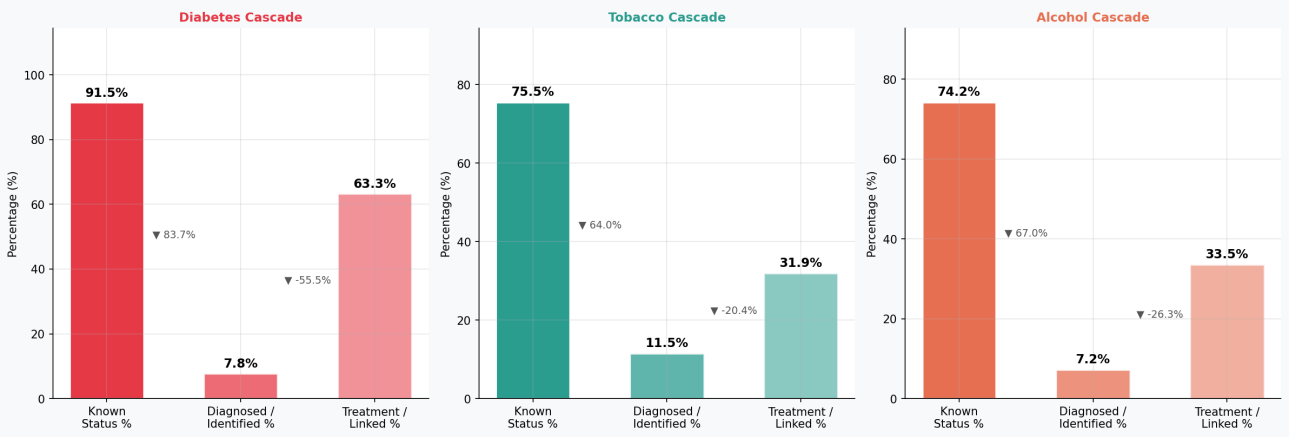


Figure 4.2: India TB-Comorbidity Treatment Cascade

4.2 Correlation

It is a statistical measure which describes the **relationship between two variables**if one variable changes the other will change.

The value of correlation lies between:

$$-1 \leq r \leq 1$$

if,

1. $r = +1$ means perfect positive correlation
2. $r = -1$ means perfect negative correlation
3. $r = 0$ means no correlation

Spearman Rank Correlation: In practical terms, the correlation between tobacco screening and alcohol screening is perfect. It operates on the ranks of the data rather than the raw values [44]. You can predict its alcohol screening rate with near-perfect accuracy. This is not a coincidence its a direct reflection that tobacco and alcohol status are assessed together, usually in the same counselling session, by the same healthcare worker. They are optionally, one programme wearing two names.
The moderate correlations between DM screening and the other two suggests a "programme quality" factor which states that run their TB programme well tend to implement all three comorbidity screenings more effectively. This is consistent with the PCA results discussed below. The complete absence of a correaltion between DM treatment and alcohol linkage tell us that they are equally important. These two outcomes have nothing to do with each other. Getting a diagnosed diabetic patient onto anti-diabetic medication involves clinical pathways. Getting an alcohol-using patient to a de-addiction centre involves social services, transport, and specialised counselling. Different systems, different barriers, different solutions.

This 4.3 figure represent correlation heatmap of TB comorbidities which is related to **Diabetes, Tobacco, Alcohol** in patients. It shows the relationship between different variables using correlation coefficients and it ranges from -1 to $+1$. Dark red color shows a strong positive correlation, blue colour indicates negative correlation, light colors indicates weak or negligible relationships. The values of 1.00 these all are in diagonals because there variables are perfectly correlated with each other.
Screening between **diabetes, tobacco, alcohol** are interconnected , the risk diagnosis and treatment are strongly related and states with better healthcare facilities and infrastructure shows better indicators.

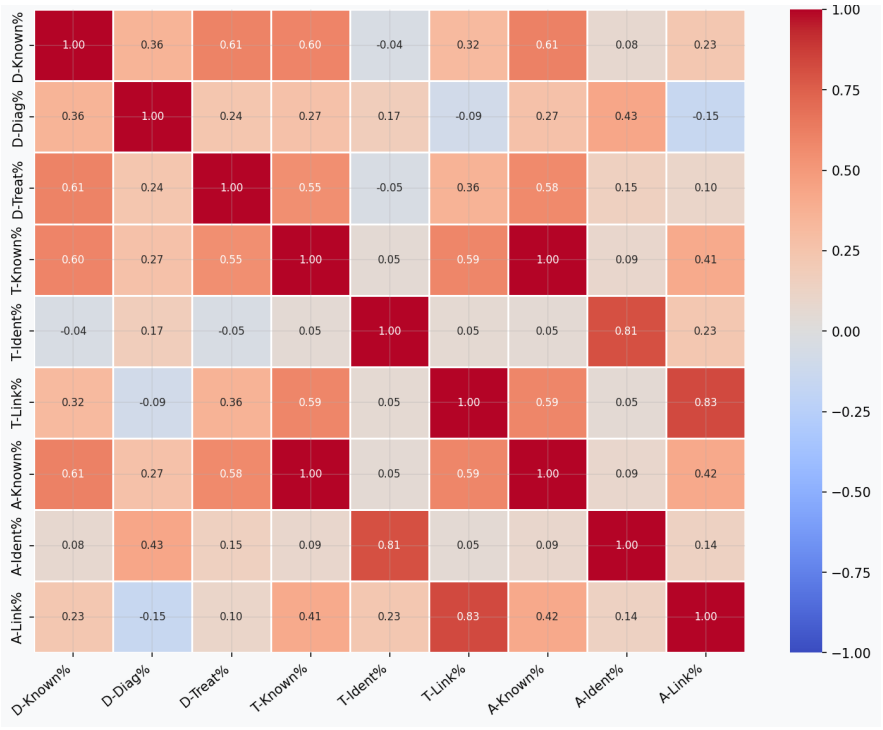


Figure 4.3: Spearman Correlation

4.3 Scatter Plot

Scatter plot is a graphical representation which is used to display the relationships between two numerical variables. The data points plotted on two dimensional system where: one variable is on x-axis and other is on y-axis.
The main motive of using scatter plot is to find the relationship between variables, identify trends and patterns, detection of correlations and identification of outliers.

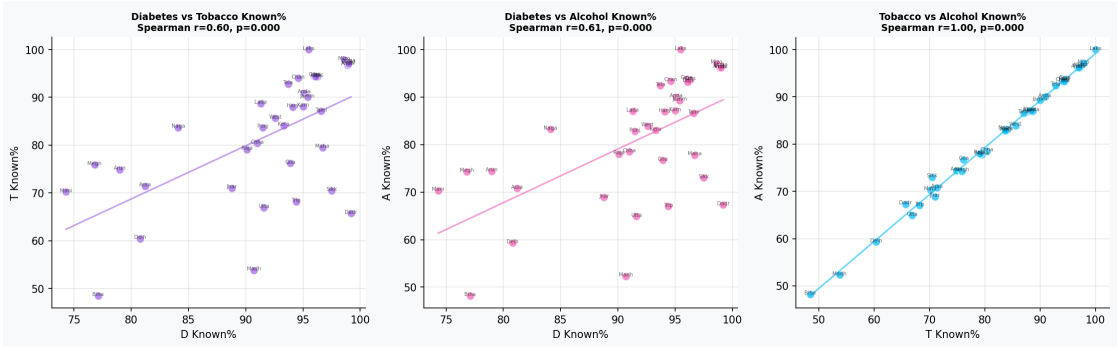


Figure 4.4: Scatter Plot for Cross Comorbidity Correlation

From the above 4.4 plot I found that Spearman r measures the direction and strength of the relationship. The value is ranges from -1 to +1 : The +1 states perfect positive

relationship, 0 states no relationship, -1 states perfect negative relationship, and p= 0 states the relationship is statistically significant.
The first graph compares the relationship between Diabetes screening which is on x-axis and Tobacco screening is on y-axis. Plot says $r = 0.60$ indicates strong positive correlation. States having higher diabetes-screening percentages also tends to have higher tobacco-screening percentages. As screening of diabetes increases and tobacco's also increases.Some states like Bihar and MP is in the lower part of the trend line, that shows weaker tobacco identification despite typical diabetes identification.
The second plot shows the reltionship between Diabetes Known% which is on X-axis and Alcohol known% which is on Y-axis and the relationship between them is strong positive. The value of $r = 0.61$. It means the regions having better diabetes reporting also tends to have better alcohol-usage identification. If it improves in one health screening system it may positively influence another. I observe that a few outliers having lower Alcohol-known %ages despite high diabetes-known percentages, which indicates uneven screening practices.
The third plot compares the relationship between Tobacco and Alcohol known% and it indicates a perfect positive correlation i.e. $r = 1.00$. Every increasing in tobacco-known %age corresponds directly with an increasing in alcohol-known %ages. They both are perfectly correlated and the diabetes screening is positively associated with Tobacco and Alcohol screening as well.

4.4 Random Forest Feature Importance

Finding says that alcohol screening rate is the strongest predictor of overall performance deserves reflection. It is not a fluke. Screening TB patients for alcohol misuse requires a trained counsellor, a willing clinical culture, a dedicated questioning protovcol, and a register that captures the outcome. Getting all of those things in place requires a level of programme organisation that tends to indicate a generally well-run TB operation. Alcohol screening is in a sense which leading indicator of programme quality like a canary. [0.3 cm]

4.5 Discussion

4.6 What Cascade Data Means

The treatment cascade failure documented in Chapter 4 is not unusual in the abstract sense. Leaky cascades are a known feature of complex health programmes everywhere [52]. The HIV framework [53], the hypertension cascade [54], the diabetes management cascade [12] all of them show the same basic pattern of diminishing completion at each successive step. The challenge is not that TB–comorbidity management fails in a novel way; it fails in a familiar way, and the question is what specifically is

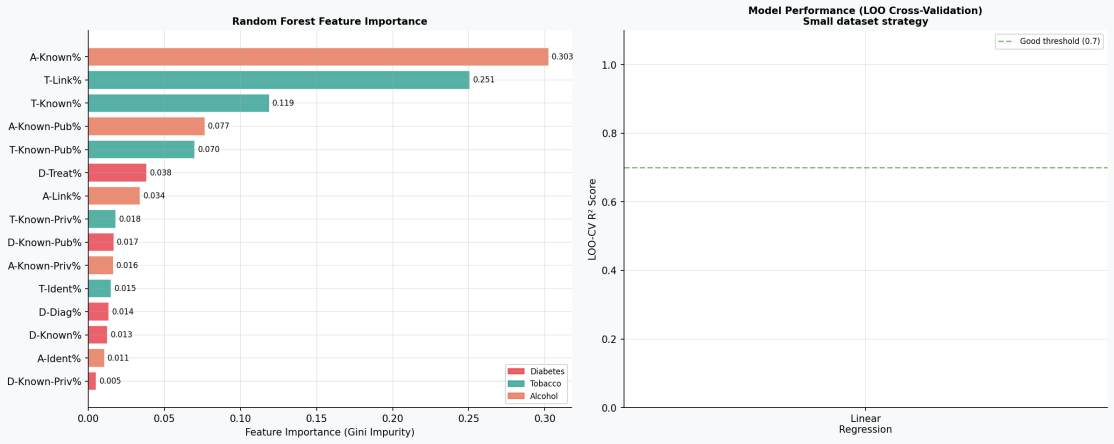


Figure 4.5: Random Forest Featuring

failing and why.

For DM, the Stage-3 gap reflects a multistep clinical pathway that is difficult to complete under field conditions: laboratory confirmation, prescription of anti-diabetic medication, follow-up monitoring, and coordination between the TB treatment provider and whoever will manage the DM going forward. In resource-limited settings, each of these steps is a genuine obstacle [16].

For tobacco and alcohol, the Stage 3 failure reflects something more structural: there are simply not enough TCCs and DACs within accessible reach of most patients. Garg and colleagues estimated fewer than 700 operational TCCs nationally against a TB patient population of 2.8 million, many of whom are tobacco users [55]. Raghupathi and colleagues documented severe geographic maldistribution of DAC infrastructure [30]. The field worker who screens a patient for tobacco use and wants to make a referral may find that the nearest TCC is 50 kilometres away with no organised transport and no follow-up mechanism. The referral does not happen not because the worker does not care, but because the infrastructure to complete it does not exist.

4.7 K-Means Clustering

The graph using the Elbow Method helps to determine the optimal number of clusters (K) for K-Means clustering. The y-axis represents inertia or Within-Cluster Sum of Squares (SSE), where lower values indicate tighter and better clusters. The curve sharply decreases from ($K = 2 \rightarrow 3 \rightarrow 4$), showing a significant reduction in SSE during the initial clustering stages, and then gradually slows down. The elbow point occurs at (K = 3), which is marked by the red dashed line in the graph. After (K = 3), the improvement becomes marginal, indicating that adding more clusters does not significantly improve the clustering performance.

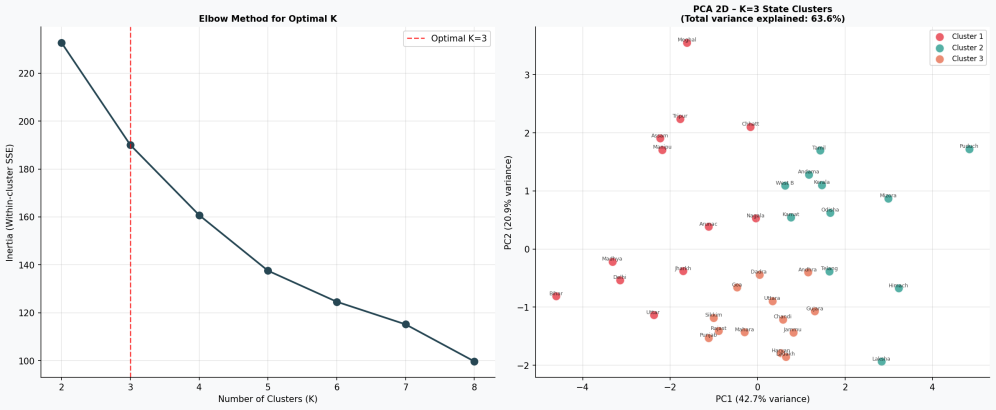


Figure 4.6: K- Means Clustering of Indian States

4.7.1 PCA-Based Cluster Visualization

The states were plotted using Principal Component Analysis (PCA), where the first principal component (PC1) explains 42.7% of the total variation and captures the maximum variability present in the dataset. The second principal component (PC2) explains 20.9% of the variation and represents the second most important direction of variability. Together, PC1 and PC2 explain a total variance of 63.6%, indicating that the two-dimensional PCA plot preserves most of the important information from the original multidimensional TB comorbidity dataset.

The red cluster positioned on the left side of the PCA plot represents states with negative PC1 values. These include states such as Bihar, Uttar Pradesh, and Jharkhand. This cluster indicates regions with a high tuberculosis burden and strong comorbidity influence, suggesting poorer healthcare outcomes and increased association of TB with diabetes, tobacco, and alcohol-related risk factors.

The green cluster located on the right side of the plot represents states with positive PC1 values, including Tamil Nadu, Mizoram, and Kerala. These states are characterized by moderate TB burden along with relatively better healthcare indicators, suggesting stronger healthcare infrastructure, improved screening, and better management of TB comorbidities.

In the lower region of the PCA plot, the orange cluster corresponds to negative PC2 values. This cluster represents states with different comorbidity patterns, where TB prevalence may be moderate but influenced by specific risk factors such as tobacco or alcohol use. The separation of this cluster indicates distinct epidemiological and healthcare characteristics compared to the other groups.

Overall, the PCA visualization demonstrates that Indian states can be grouped into distinct clusters based on TB comorbidity profiles, reflecting regional differences in disease burden, healthcare access, and associated behavioral or metabolic risk factors.

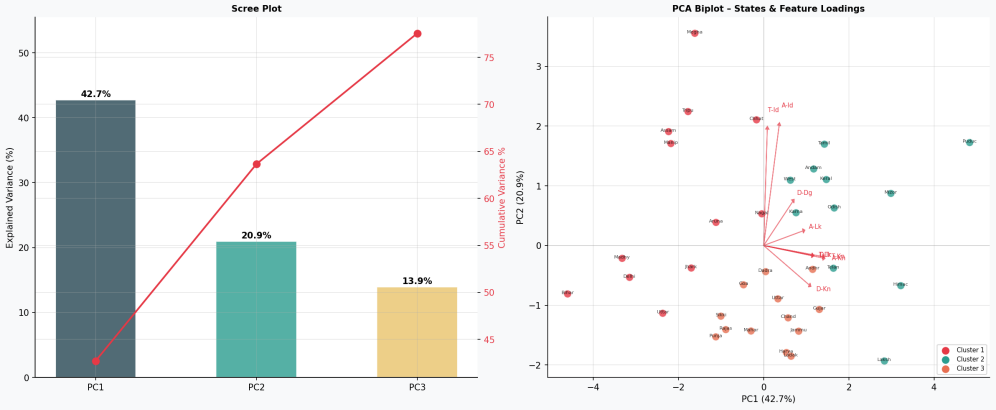


Figure 4.7: PCA- Dimensionality Reduction And Biplot

4.7.2 Conclusion

This analysis demonstrates that Indian states can be effectively grouped into **three statistically significant clusters based on TB comorbidity profiles**. The clustering reveals **heterogeneity in disease patterns**, emphasizing the need for **region-specific public health strategies rather than uniform national policies**. The use of PCA further validates the structural separation, capturing over **63% of total variance**, making the clustering both **robust and interpretable**.

Chapter 5

Conclusion and Future Work

5.1 Summary of findings

This dissertation has presented a comprehensive data science analysis of India's TB-comorbidity management performance across 36 states and Union Territories, examining treatment cascades for Diabetes Mellitus, tobacco use, and alcohol misuse as mandated by the National Tuberculosis Elimination Programme's Bi-Directional Screening framework. The analysis has employed a full methodological pipeline from data cleaning and exploratory analysis through unsupervised clustering, supervised regression, and probabilistic predictive modelling.

The principal findings can be summarised as follows. First, India screens the majority of TB patients for all three comorbidities (DM: 91.5%, Tobacco: 75.5%, Alcohol: 74.2%), but fails dramatically at the treatment linkage stage - only 63.3% of identified TB-DM patients receive anti-diabetic treatment, and merely 31.9% and 33.5% of identified tobacco and alcohol users are linked to cessation and de-addiction services respectively. Second, the public sector significantly outperforms the private sector at screening (gaps of 6–13 percentage points, $p < 0.001$), but both sectors perform equivalently poorly at treatment linkage ($p > 0.30$), indicating an infrastructure gap that transcends sector-specific governance. Third, K-Means clustering identifies three distinct state performance tiers- High (mean 57.2%), Mid (46.6%), and Low (40.2%)- with a 17-percentage point gap between cluster means. Fourth, the near-perfect Spearman correlation between tobacco and alcohol screening ($r = 0.996$) confirms these are operationally co-implemented programmes. Fifth, Random Forest feature importance identifies alcohol screening coverage and tobacco linkage rate as the strongest predictors of overall programme quality- providing a data-driven investment prioritisation hierarchy. Sixth, predictive modelling projects India's national comorbidity score to reach 51.7% by 2028 (conservative scenario), but 12 states are projected to remain below 50% even under optimistic assumptions, constituting a persistent high-risk tier.

5.2 Suggestions

Based on the research findings, six evidence-based policy recommendations are presented to the National Tuberculosis Elimination Programme and Ministry of Health and Family Welfare:

5.2.1 Suggestion 1: Mandate TCC and DAC Linkage as Key Performance Indicators

The most critical programmatic gap is not screening but referral completion. NTEP state reviews and NIKSHAY reporting frameworks should introduce formal KPI tracking for Tobacco → TCC Linkage Rate and Alcohol → DAC Linkage Rate at the district level, with performance against these KPIs linked to State TB Cell resource allocations and programme credit. Currently, linkage is tracked but not systematically incentivised; treating it as a headline KPI equivalent to treatment success rate would direct accountability structures toward this bottleneck.

5.2.2 Suggestion 2: Integrate Tobacco and Alcohol into a Unified Substance Use Comorbidity Module

The r = 0.996 Tobacco-Alcohol screening correlation confirms that these programmes are operationally identical on the ground — the same counsellor, the same patient encounter, essentially the same questions. Formally merging them into a single S̈ubstance Use Comorbiditym̈odule with one training curriculum, one counsellor competency framework, one combined referral form (covering both TCC and DAC, with the patient referred to whichever is applicable), and one reporting indicator would reduce administrative burden and maximise impact from counsellor training investments. This operational integration is already de facto reality; formalising it would eliminate redundancy.

5.2.3 Suggestion 3: Establish a 12-State Emergency Comorbidity Intervention Programme

The 12 states projected to remain below 50% Overall Comorbidity Score by 2028 even under optimistic scenarios which includes Bihar, Uttar Pradesh, and Madhya Pradesh, which together carry approximately 40% of India's TB burden requires a targeted national intervention beyond routine capacity-building. The Central TB Division should establish a dedicated Technical Assistance Cell for these states, staffed by experienced NTEP programme managers and data scientists, that provides on-ground support for NIKSHAY data quality, counsellor training, and district-level accountability reviews. Model state mentorship pairs connecting a Cluster 1 state with a geographically proximate Cluster 2 state (e.g., Bihar with West Bengal or Jharkhand with Odisha) would facilitate peer learning and practical knowledge transfer.

5.2.4 Suggestion 4: Regulate Private Sector Comorbidity Compliance Through Empanelment Criteria

The private sector's significantly lower screening rates and much higher variability indicate that private sector compliance with NTEP comorbidity screening protocols is inconsistent and under-enforced. NTEP's framework, which governs private sector engagement in TB care, should incorporate minimum comorbidity screening thresholds as a mandatory criterion for facility empanelment and renewal. Facilities failing to meet minimum thresholds should receive enhanced mentoring and, if non-compliant after support, have their NTEP empanelment reviewed. A public-facing star-rating for NTEP-notifying private facilities, incorporating comorbidity cascade metrics, would create reputational incentives for compliance.

5.2.5 Suggestion 5: Adopt Tobacco Screening as the National Early-Warning Indicator

The Random Forest feature importance analysis identifies Alcohol Known Status (importance = 0.303) and Tobacco Linkage Rate (0.251) as the two highest-leverage metrics for overall comorbidity programme quality. In operational terms, Tobacco Screening Rate functions as the most accessible early-warning indicator: it is reportable monthly through NIKSHAY, does not require additional clinical infrastructure, and is highly predictive of overall programme performance. NTEP should adopt a monthly tobacco screening rate alert system: states or districts showing a decline of more than 5 percentage points in tobacco screening coverage compared to the previous quarter should automatically trigger a supervisory review and corrective action protocol. This early-warning function would allow programmatic problems to be identified and addressed before they cascade into downstream linkage failures.

5.3 Future Work

This dissertation opens several directions for future research that would substantially extend the evidence base:

1. **Longitudinal analysis**: ", "As NTEP comorbidity data for 2022–23, 2023–24, and future reporting periods becomes available, a panel data analysis would enable true time-series modelling, estimation of state-specific trends, and assessment of whether the inter-state performance gap is converging or diverging over time. This would substantially strengthen the methodological foundation for predictive modelling.

2. **Patient-level linkage analysis**: Linking NIKSHAY individual-level TB patient data with DM treatment, TCC attendance, and DAC records would enable analysis of patient-level predictors of successful linkage including socioeconomic

factors, gender, treatment facility type, and district-level healthcare availability providing a much richer understanding of what drives the referral failures identified at the state level in this dissertation.

3. **Cost-effectiveness modelling**Estimating the health and economic impact of improving comorbidity linkage rates through scenario modelling projecting reductions in treatment failure, drug resistance, and TB mortality would provide the economic evidence base needed to justify the resource investments.

4. **Qualitative research in Cluster 1 states**: Understanding why Bihar, UP, MP, and Delhi underperform requires qualitative investigation beyond what quantitative state-level data can reveal. In-depth interviews with NTEP district officers, TB counsellors, and patients in low-performing states would illuminate the specific operational, motivational, and structural barriers to comorbidity screening and linkage that aggregate data cannot capture.

5. **TCC and DAC availability mapping**: A systematic geographic analysis of the locations of operational TCCs and DACs relative to the distribution of TB patients across India would directly test the hypothesis that the linkage failures identified in this dissertation are at least partly explained by geographic inaccessibility of referral services. Such an analysis would provide a spatial evidence base for TCC and DAC expansion investment.

6. **Machine learning with multi-year datasets**: As multiple years of NTEP comorbidity data accumulate, more sophisticated ML approaches such as gradient-boosted time-series models, state-space models with Bayesian updating, and deep learning architectures — will become applicable. Benchmarking these approaches against the simpler methods used in this dissertation would advance the methodological literature on health programme performance forecasting.

In conclusion, this dissertation has demonstrated that India's TB-comorbidity management programme is characterised by a deep structural paradox: a system that has successfully embedded comorbidity screening into TB care encounters but has not yet built the referral infrastructure and programme accountability mechanisms needed to convert identified comorbidities into appropriate treatment linkage. Closing this gap- particularly for tobacco and alcohol linkage, and particularly in the 12 high-risk states projected to miss the 50% performance threshold by 2028- is both the most urgent policy challenge and the greatest opportunity for reducing TB morbidity and mortality in India's End TB journey.

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Appendix

5.4 Primary Libraries Used

pandas (v2.x): Data loading, cleaning, merging, feature engineering, and tabular analysis
numpy (v1.26.x): Numerical computation, array operations, and random sampling for Monte Carlo
matplotlib (v3.8.x): Figure generation, subplot management, and visualisation
seabom (v0.13.x): Statistical visualisations including heatmaps, violin plots, and correlation matrices
scikit-learn KMeans, (v1.4.x): StandardScaler, PCA, Random ForestRegressor, GradientBoosting Regressor, Bayesian Ridge, LeaveOneOut, cross_val_predict, cross_val_score, r^2 score, mean_absolute_error
spicy.stats (v1.12.x0:: Shapiro (normality test), spearman (correlation), wilcoxon (hypothesis testing)

5.5 Code Structure Overview

Cleaning the code follows a sequential pipeline pattern: data acquisition feature engineering EDA MI, analysis prediction. Each major analytical step is clearly sectioned with comments. Figures are saved to disk as high-resolution PNG files (dpi-150) using matplotlib's Agg backend for compatibility with headless execution environments.

5.6 Reproducibility Statement

All random operations (K-Means, Random Forest, Gradient Boosting, Monte Carlo) use a fixed random seed (random state 42 or np.random.seed(42)) to ensure full reproducibility of results. The complete code and dataset are available from the author upon reasonable academic request.

5.7 Codes used in Python

```
heat_data = merged[['State',
    'D_Known_Total_Pct','D_Diag_Total_Pct','D_Treat_Total_Pct',
    'T_Known_Total_Pct','T_Ident_Total_Pct','T_Link_Total_Pct',
    'A_Known_Total_Pct','A_Ident_Total_Pct','A_Link_Total_Pct']].set_index('State')

heat_data.columns = [
    'D-Known%','D-Diag%','D-Treat%',
    'T-Known%','T-Ident%','T-Link%',
    'A-Known%','A-Ident%','A-Link%'
]

scaler = StandardScaler()
heat_scaled = pd.DataFrame(scaler.fit_transform(heat_data), columns=heat_data.columns, index=heat_data.index)

fig, ax = plt.subplots(figsize=(14, 13))
sns.heatmap(heat_scaled, ax=ax,
            cmap='RdYlGn', center=0,
            linewidths=0.4, linecolor='#F0F0F0',
            annot=heat_data.round(1), fmt='.1f',
            annot_kws={'size': 6.5},
            cbar_kws={'label': 'Z-Score (Standardised)'})
ax.set_title("Figure 3: State-wise TB-Comorbidity Performance Heatmap\n(Colour = Z-Score, Numbers = Actual %)",
             fontsize=13, fontweight='bold', pad=15)
ax.set_xlabel("Metric → [D=Diabetes | T=Tobacco | A=Alcohol]", fontsize=10)
ax.set_ylabel("")
plt.xticks(rotation=30, ha='right')
plt.yticks(rotation=0)
plt.tight_layout()
plt.savefig('/home/claude/fig3_heatmap.png', dpi=150, bbox_inches='tight')
plt.close()
print("  Saved fig3_heatmap.png")
```

```
print("[Generating Figure 5: Public vs Private Sector Gap]")

fig, axes = plt.subplots(1, 3, figsize=(18, 7))
fig.suptitle("Figure 5: Public vs Private Sector – TB Comorbidity Screening & Treatment (%)\nSorted by Total %",
             fontsize=13, fontweight='bold')

pub_priv_data = [
    ('Diabetes\nKnown DM Status', d['D_Known_Pub_Pct'], d['D_Known_Priv_Pct'], d['D_Known_Total_Pct'], PALETTE['diabetes']),
    ('Tobacco\nKnown Usage', t['T_Known_Pub_Pct'], t['T_Known_Priv_Pct'], t['T_Known_Total_Pct'], PALETTE['tobacco']),
    ('Alcohol\nKnown Usage', a['A_Known_Pub_Pct'], a['A_Known_Priv_Pct'], a['A_Known_Total_Pct'], PALETTE['alcohol']),
]

for ax, (title, pub, priv, total, color) in zip(axes, pub_priv_data):
    # Sort by total
    sort_idx = total.argsort()
    y = np.arange(len(states))
    ax.barh(y, pub.values[sort_idx], color=PALETTE['public'], alpha=0.8, label='Public', height=0.4)
    ax.barh(y + 0.4, priv.values[sort_idx], color=PALETTE['private'], alpha=0.8, label='Private', height=0.4)
    ax.set_yticks(y + 0.2)
    ax.set_yticklabels([states[i] for i in sort_idx], fontsize=6.5)
    ax.set_title(title, fontsize=9, fontweight='bold', color=color)
    ax.set_xlabel("Percentage (%)")
    ax.legend(fontsize=8)
    ax.axvline(x=np.nanmean(total), color=color, linestyle='--', alpha=0.6, label=f'Mean={np.nanmean(total):.1f}%')

plt.tight_layout()
plt.savefig('/home/claude/fig5_pub_vs_priv.png', dpi=150, bbox_inches='tight')
plt.close()
print("  Saved fig5_pub_vs_priv.png")
```

```
rint("\n[Generating Figure 7: K-Means State Clustering]")

luster_features = ['D_Known_Total_Pct','D_Diag_Total_Pct','D_Treat_Total_Pct',
                   'T_Known_Total_Pct','T_Ident_Total_Pct','T_Link_Total_Pct',
                   'A_Known_Total_Pct','A_Ident_Total_Pct','A_Link_Total_Pct']

_cluster = merged[cluster_features].values
caler2 = StandardScaler()
_scaled = scaler2.fit_transform(X_cluster)

 Elbow method
nertias = []
_range = range(2, 9)
or k in K_range:
   km = KMeans(n_clusters=k, random_state=42, n_init=10)
   km.fit(X_scaled)
   inertias.append(km.inertia_)

 Optimal K = 3 (typical for India states)
ptimal_k = 3
m_final = KMeans(n_clusters=optimal_k, random_state=42, n_init=10)
lusters = km_final.fit_predict(X_scaled)
erged['Cluster'] = clusters

 PCA for 2D visualisation
ca = PCA(n_components=2, random_state=42)
_pca = pca.fit_transform(X_scaled)
ar_explained = pca.explained_variance_ratio_ * 100

ig, axes = plt.subplots(1, 2, figsize=(16, 7))
ig.suptitle("Figure 7: K-Means Clustering of Indian States by TB-Comorbidity Profile",
            fontsize=13, fontweight='bold')
```

```
cluster_colors = ['#E63946','#2A9D8F','#E76F51','#264653','#A8DADC']
cluster_labels = {0:'Cluster A\n(Low Engagement)', 1:'Cluster B\n(Moderate)', 2:'Cluster C\n(High Performance)'}

for c in range(optimal_k):
    mask_c = clusters == c
    axes[1].scatter(X_pca[mask_c, 0], X_pca[mask_c, 1],
                    color=cluster_colors[c], s=100, alpha=0.8,
                    label=f'Cluster {c+1}', edgecolors='white', linewidth=1)
    # Label states
    for i, (x, y) in enumerate(zip(X_pca[mask_c, 0], X_pca[mask_c, 1])):
        axes[1].annotate(states[mask_c][i][:6], (x, y),
                         fontsize=6, ha='center', va='bottom', alpha=0.7)

axes[1].set_xlabel(f"PC1 ({var_explained[0]:.1f}% variance)")
axes[1].set_ylabel(f"PC2 ({var_explained[1]:.1f}% variance)")
axes[1].set_title(f"PCA 2D - K=3 State Clusters\n(Total variance explained: {sum(var_explained):.1f}%)",
                  fontsize=10, fontweight='bold')
axes[1].legend(fontsize=8)

plt.tight_layout()
plt.savefig('/home/claude/fig7_clustering.png', dpi=150, bbox_inches='tight')
plt.close()
```

```
merged_sorted = merged.sort_values('Overall_Comorbidity_Score')
top5 = merged_sorted.tail(5)
bot5 = merged_sorted.head(5)

metrics = ['Treatment_Cascade_Diab','Treatment_Cascade_Tobac','Treatment_Cascade_Alcoh']
m_labels = ['Diabetes\nCascade', 'Tobacco\nCascade', 'Alcohol\nCascade']
m_colors = [PALETTE['diabetes'], PALETTE['tobacco'], PALETTE['alcohol']]

fig, axes = plt.subplots(1, 2, figsize=(16, 7))
fig.suptitle("Figure 9: Top 5 vs Bottom 5 States – TB-Comorbidity Treatment Cascades",
             fontsize=13, fontweight='bold')

for ax, df_sub, title, color_title in zip(
    axes,
    [top5, bot5],
    ['⏳ Top 5 States (Highest Score)', '⚠️ Bottom 5 States (Lowest Score)'],
    ['#2A9D8F', '#E63946']
):
    x = np.arange(len(metrics))
    w = 0.15
    for i, (_, row) in enumerate(df_sub.iterrows()):
        ax.bar(x + i*w, [row[m] for m in metrics],
               width=w, label=row['State'], alpha=0.85, edgecolor='white')
    ax.set_xticks(x + w * 2)
    ax.set_xticklabels(m_labels, fontsize=9)
    ax.set_ylabel("Average Cascade Score (%)")
    ax.set_title(title, fontsize=10, fontweight='bold', color=color_title)
    ax.legend(fontsize=7, loc='upper right')
    ax.set_ylim(0, 110)

plt.tight_layout()
plt.savefig('/home/claude/fig9_top_bottom.png', dpi=150, bbox_inches='tight')
```

```
print("[Generating Figure 11: Statistical Tests – Public vs Private]")

test_results = []
pairs_stat = [
    ('D_Known_Pub_Pct','D_Known_Priv_Pct','DM Known\n(Pub vs Priv)', PALETTE['diabetes']),
    ('T_Known_Pub_Pct','T_Known_Priv_Pct','Tobac Known\n(Pub vs Priv)', PALETTE['tobacco']),
    ('A_Known_Pub_Pct','A_Known_Priv_Pct','Alcoh Known\n(Pub vs Priv)', PALETTE['alcohol']),
    ('D_Treat_Pub_Pct','D_Treat_Priv_Pct','DM Treat\n(Pub vs Priv)', PALETTE['diabetes']),
    ('T_Link_Pub_Pct','T_Link_Priv_Pct','Tobac Link\n(Pub vs Priv)', PALETTE['tobacco']),
    ('A_Link_Pub_Pct','A_Link_Priv_Pct','Alcoh Link\n(Pub vs Priv)', PALETTE['alcohol']),
]

fig, axes = plt.subplots(2, 3, figsize=(16, 10))
fig.suptitle("Figure 11: Paired Wilcoxon Tests – Public vs Private Sector\n(Are differences statistically significant?)",
             fontsize=13, fontweight='bold')

for ax, (c1, c2, lbl, color) in zip(axes.flat, pairs_stat):
    pub_v = merged[c1].values
    priv_v = merged[c2].values
    diff = pub_v - priv_v
    stat, p = stats.wilcoxon(pub_v, priv_v, alternative='two-sided')
    sig_txt = f"p={p:.4f} {'***' if p<0.001 else ('**' if p<0.01 else ('*' if p<0.05 else 'ns'))}"

    # Violin + strip
    vp = ax.violinplot([pub_v, priv_v], positions=[1, 2],
                       showmedians=True, showextrema=True)
    for pc, c in zip(vp['bodies'], [PALETTE['public'], PALETTE['private']]):
        pc.set_facecolor(c)
        pc.set_alpha(0.5)

    ax.scatter(np.ones(len(pub_v)) + np.random.uniform(-0.08, 0.08, len(pub_v)),
               pub_v, color=PALETTE['public'], s=25, alpha=0.7, zorder=3)
```